

Project Design Phase-II Technology Stack (Architecture & Stack)

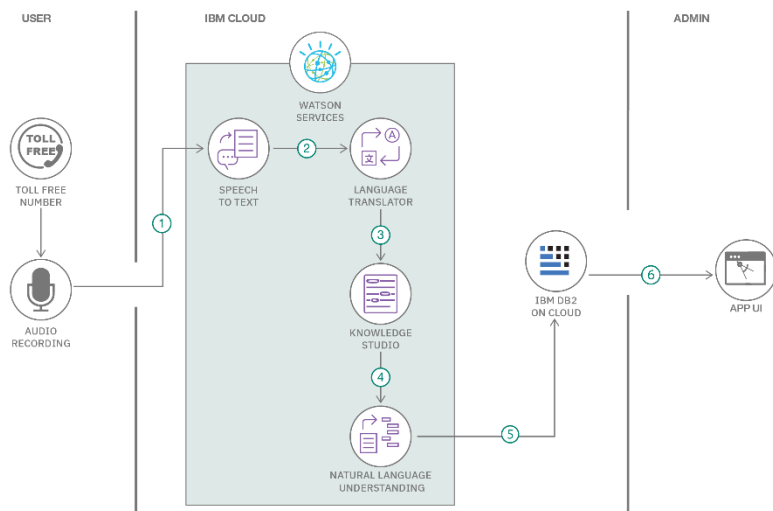
Date	31 January 3035
Team ID	
Project Name	
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Customer and Admin Web Application	React.js, Tailwind CSS, shadcn/ui
2.	Application Logic	Business Logic	Node.js, Express.js
3.	Authentication	User Authentication	JWT, bcryptjs
4.	Database	Data Storage	MongoDB, Mongoose
5.	File Storage	Product Images	Cloudinary
6.	Payment API	Online Payments	Razorpay
7.	Notifications	Toast Messages	React Hot Toast
8.	Charts	Dashboard Analytics	Chart.js
9.	Version Control	Source Code	Git, GitHub
10. 0	Infrastructure	Deployment	Vercel, Render, MongoDB Atlas

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frontend and Backend frameworks	React, Express, Tailwind CSS
2.	Security Implementations	Authentication and authorization	JWT, bcryptjs, Helmet, CORS
3.	Scalable Architecture	Three-tier MVC architecture	MERN Stack

S.No	Characteristics	Description	Technology
4.	Availability	Cloud deployment	Vercel, Render, MongoDB Atlas
5.	Performance	Optimized APIs and responsive UI	Axios, Mongoose Indexing, Lazy Loading

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>